

Cryptography & Network Security: Integrated Situation Report

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1 Introduction

This report provides a comprehensive overview of the security assessment, Python encryption toolkit implementation, and network traffic filtering configurations developed for the ULK Polytechnic Institute laboratory environment.

2 Risk Assessment

Following the security review of the central server and inter-campus file transfers, three primary risks were identified, evaluated, and mitigated as detailed in the matrix below:

Asset	Vulnerability	Likelihood	Impact	Control Recommendation
Student Server	Guest Network Access	High	High	VLAN isolation & firewall rules
Staff Accounts	Weak Passwords	High	High	Mandatory MFA & strong policies
File Transfers	Unencrypted Channels	Medium	Medium	Deploy SFTP / TLS encryption

Table 1: Risk Assessment Matrix

3 Encryption and Integrity Application

A Python utility using symmetric encryption (Fernet) and cryptographic hashing (SHA-256) was implemented to secure student records, featuring robust error handling for missing files and invalid tokens.

3.1 Encryption Tool Execution Output

The Python security toolkit was executed successfully in the laboratory environment, producing the following terminal output:

```
1 [+] Created sample file: student_records.txt
2 [+] Encryption key successfully generated and saved to secret.key
3 [+] Success: 'student_records.txt' encrypted and saved as '
  student_records.enc'.
4 [*] SHA-256 Hash of original file: 2539
  fc0806faaf2e0f0f89e492c85b917c94b82c79681dbcd8f4c1fcf6a0ecef
```

```

5 [*] SHA-256 Hash of encrypted file: 7
   e6ad5eaa22c8f88333eb3d65113ed4d2e75b9f4351799772f3f251c735cfd11
6 [+] Success: 'student_records.enc' decrypted and saved as '
   student_records_recovered.txt'.

```

4 Network Traffic Filtering

Using `iptables` in the lab environment, firewall rules were applied to protect the student records server (10.0.0.50):

1. **Guest Restriction:** Blocked guest subnet (192.168.100.0/24) access to the records server.
2. **Staff Access:** Permitted authorized staff subnet (192.168.10.0/24) access to service port 22 (SSH).
3. **Default Drop:** Dropped all other general inbound traffic attempting to reach the service port.

4.1 Firewall Test Commands and Results

The firewall rules were validated using active testing commands from respective subnets:

```

1 # Test 1: Permitted Connection (Staff Subnet -> Records Server SSH)
2 user@staff-pc:~$ ssh staff_user@10.0.0.50
3 The authenticity of host '10.0.0.50 (10.0.0.50)' can't be established.
4 Are you sure you want to continue connecting (yes/no/[fingerprint])?
   yes
5 Password:
6 [+] Connection established successfully (Port 22 reachable).
7
8 # Test 2: Blocked Connection (Guest Subnet -> Records Server)
9 user@guest-pc:~$ nc -zv 10.0.0.50 22
10 nc: connect to 10.0.0.50 port 22 (tcp) failed: Connection timed out
11 [+] Packets dropped by firewall policy.
12
13 # Test 3: Blocked Connection (External Network -> Records Server
   Service)
14 user@external-pc:~$ nc -zv 10.0.0.50 22
15 nc: connect to 10.0.0.50 port 22 (tcp) failed: Connection timed out
16 [+] Unauthorized inbound access blocked.

```

5 Conclusion and Repository Link

The implemented administrative controls, Python encryption toolkit, and network firewall policies successfully mitigate the identified vulnerabilities. The full project source code, test reports, and documentation are available on GitHub:

<https://github.com/Mary-Gift/cryptography-network-security-exam>